SG No. 54 D_{2h}^{8} $Pcca$ [orthorhombic]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `4a`, site symmetry: `-1`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `4b`, site symmetry: `-1`

Table 2: Wyckoff bond: `4a@4b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-5]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-6]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-7]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-8]`

* Wyckoff site: `4c`, site symmetry: `.2.`

Table 3: Wyckoff bond: `4a@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, y, \frac{1}{4}\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[0, -y, \frac{3}{4}\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[6,-8]`

Table 4: Wyckoff bond: `4b@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[0, y, \frac{1}{4}\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[2,4]`
3	$[0, -Y, 0]$	$\left[0, -y, \frac{3}{4}\right]$	`[5,7]`
4	$[0, Y, 0]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[6,8]`

Table 5: Wyckoff bond: `8c@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, y, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, y, \frac{1}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[4]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -Y, -Z]$	$\left[0, -y, \frac{3}{4}\right]$	[5]
6	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	[6]
7	$[X, -Y, Z]$	$\left[0, -y, \frac{3}{4}\right]$	[7]
8	$[-X, Y, Z]$	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	[8]

* Wyckoff site: 4d, site symmetry: ..2

Table 6: Wyckoff bond: 4a@4d, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, 0, z\right]$	[1,-2]
2	$[-X, Y, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	[3,-4]
3	$[-X, -Y, 0]$	$\left[\frac{3}{4}, 0, -z\right]$	[5,-6]
4	$[X, -Y, 0]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	[7,-8]

Table 7: Wyckoff bond: 4b@4d, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, 0, z\right]$	[1,2]
2	$[0, 0, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	[3,4]
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	[5,6]
4	$[0, 0, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	[7,8]

Table 8: Wyckoff bond: 8c@4d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, 0, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, 0, z\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2} - z\right]$	[4]
5	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	[5]
6	$[X, Y, -Z]$	$\left[\frac{3}{4}, 0, -z\right]$	[6]
7	$[X, -Y, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	[7]
8	$[-X, Y, Z]$	$\left[\frac{1}{4}, 0, z + \frac{1}{2}\right]$	[8]

* Wyckoff site: 4e, site symmetry: ..2

Table 9: Wyckoff bond: 4a@4e, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	[1,-2]
2	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2} - z\right]$	[3,-4]
3	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	[5,-6]
4	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, z + \frac{1}{2}\right]$	[7,-8]

Table 10: Wyckoff bond: `4b@4e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$[0,\ 0,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$[0,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 11: Wyckoff bond: `8c@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[8]`

* Wyckoff site: `8f`, site symmetry: `1`

Table 12: Wyckoff bond: `8a@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y,\ -z\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	`[8]`